METRIC
MIL-DTL-32632
25 July 2019

DETAIL SPECIFICATION

VALVES, FLOW CONTROL, DILATING DISK, FOR NAVAL SHIPBOARD WATER, OIL, AND GAS SERVICE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flow control valves of dilating, low torque, hysteresis-free design employing multiple-segment irising seat-disks construction for naval shipboard water, oil, and gas services. These valves are for precise flow control of liquids and gases and allow unconstrained flow when fully open.

1.2 Classification. Valves are either unclassified or classified in accordance with ASME B16.34 (see 1.3) with the end type, size, operator to be used and potable water suitability as specified (see 6.2).

1.3 Part or identifying number (PIN). The PIN to be used for parts acquired to this specification is constructed as follows:

M	**32632**	**-**	**XXXX**	**X**	**-**	**XXX**	**X**	**XX**
Prefix for Military Specification	Specification Number	-	Classification (see 1.2 and code below)	End Type (see code below)	-	Size (see code below)	Operator (see code below)	Potable Water Use (see code below)
Example: M32632-0150F-080HPW is for an ASME class 150, flanged end, Diamètre Nominal (DN) 80 (Nominal Pipe Size [NPS] 3), handwheel operated, potable water suitable, dilating flow control valve.								

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4820

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Classification	Code	End Type	Code	NPS	Code (DN Size)	Operator	Code	Potable Water Use	Code
Unclassified	0000	Welded	W	¼	008	Lever	L	Suitable	PW
ASME 150	0150	ASME Flanged	F	1	025	Handwheel	H	Not Suitable	(blank)
ASME 300	0300	Navy Flanged	N	1½	040	Electric Actuator	E		
ASME 600	0600			2	050	Other	T		
ASME 900	0900			3	080				
ASME 1500	1500			4	100				
				6	150				
				8	200				
				10	250				
				12	300				
				14	350				
				16	400				

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications where a High Degree of Reliability is Required; General Specification for

DOD-V-24657 - Valve Actuator, Direct Coupled, Gear Driven Electrically Powered for Naval Shipboard Propulsion and Auxiliary Systems (Metric)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

MIL-STD-798 - Design Evaluation Requirements for Valves for Naval Shipboard Use

MIL-STD-1472 - Human Engineering

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

803-1385620 - Handwheels for Valves

803-1385892 - Flanges, Bronze, Special, for Use with Butterfly Valves

(Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Technical Data Management Information System [TDMIS] at https://mercury.tdmis.navy.mil/. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository [NSEDR] online at https://199.208.213.105/webjedmics/index.jsp. To request an NSEDR account for drawing access, send an email to NNSY_JEDMICS_NSEDR_HELP_DESK@navy.mil.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

S9086-RK-STM-010/505 - Naval Ships' Technical Manual, Chapter 505, Piping Systems

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of these documents are available online via Technical Data Management Information System [TDMIS] at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering [hard copy] via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

ANSI/FCI 70-2 - Control Valve Seat Leakage

ANSI/ISA 75.02 - Control Valve Capacity Test Procedures

(Copies of these documents are available online at https://webstore.ansi.org.)

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 - Unified Inch Screw Threads, (UN and UNR Thread Form)

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ Through NPS 24 Metric/Inch Standard

ASME B16.10 - Face-to-Face and End-to-End Dimensions of Valves

ASME B16.34 - Valves Flanged, Threaded, and Welding End

ASME B18.2.1 - Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

ASME B18.2.2 - Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of these documents are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A182/A182M - Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

ASTM A240/A240M - Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A351/A351M - Standard Specification for Castings, Austenitic, for Pressure-Containing Parts

ASTM A479/A479M - Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels

ASTM A564/A564M - Standard Specification for Hot-Rolled and Cold-Finished Age-Hardening Stainless Steel Bars and Shapes

ASTM A743/A743M - Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel, Corrosion Resistant, for General Application

ASTM B61 - Standard Specification for Steam or Valve Bronze Castings

ASTM D2000 - Standard Classification System for Rubber Products in Automotive Applications

ASTM D4894 - Standard Specification for Polytetrafluoroethylene (PTFE) Granular Molding and Ram Extrusion Materials

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 6708 - Pipework components -- Definition and selection of DN (nominal size)

(Copies of this document are available online at www.iso.org.)

NSF INTERNATIONAL/ANSI

NSF/ANSI 61 - Drinking Water Systems Components - Health Effects

(Copies of this document are available online at www.nsf.org.)

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SAE INTERNATIONAL

SAE AMS7259 - Rubber: Fluorocarbon (FKM), High Temperature/Fluid Resistant, Low Compression Set/ 85 to 95 Hardness, for Seals in Fuel Systems and Specific Engine Oil Systems

SAE AMS7276 - Rubber: Fluorocarbon (FKM) High-Temperature-Fluid Resistant Low Compression Set for Seals in Fuel Systems and Specific Engine Oil Systems

SAE J1086 - Numbering Metals and Alloys

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Materials. The materials for dilating flow control valves shall be as specified herein. Materials for components not specifically detailed herein shall be selected to prevent galling, seizing, and excessive wear of operating parts and galvanic compatibility with system piping.

3.2.1 Metallic parts. Unless otherwise specified (see 6.2), the bodies, covers, and base plates shall be made from materials as specified in a through e below. Sector leaves, gears, pinions, assembly hardware, and other components shall be as specified in 3.2.1.f below.

a. ASTM A182/A182M, grade F304 (UNS S30400) 1/ or grade F316 (UNS S31600) or

b. ASTM A479/A479M, type 304 or

c. ASTM A351/A351M, grade CF3 (UNS J92500) or grade CF3M (UNS J92800) or

d. ASTM A743/A743M, grade CF16F (UNS J92701 or S30300) or

e. ASTM A240/A240M, grade F329 (UNS S32900) or

f. Other materials 2/ as specified (see 6.2) appropriate to the temperature, pressure, and fluid compatibility of the valve, such as valve bronze in accordance with ASTM B61. Parts made from ASTM A564/A564M UNS S17400 shall be heat treated to condition H1100.

(Note 1/: The SAE J1086 Unified Numbering System (UNS) designation permits the use of any one of the many alloy specifications and standards listed for that UNS number.)

(Note 2/: Valves for seawater service shall be provided with nickel-copper alloy trim, such as discs, seats, and stem guides as applicable.)

3.2.2 Fasteners. Unless otherwise specified (see 6.2), fasteners shall be in accordance with MIL-DTL-1222. Threads shall be in accordance with ASME B1.1. Bolts and nuts shall be in accordance with ASME B18.2.1 and ASME B18.2.2.

3.2.3 Non-metallic parts. Unless otherwise specified (see 6.2), seals, seats, gaskets, diaphragms, bearings, and other non-metallic parts shall be commercial and made of the following:

a. Acrylonitrile butadiene (Buna N) in accordance with ASTM D2000 BF, BG, or BK; or

b. Ethylene propylene diene terpolymer (EPDM) in accordance with ASTM D2000 DA; or

c. Fluorocarbon elastomer (FKM) in accordance with SAE AMS7276 or SAE AMS7259; or

d. Polytetraflouroethylene (PTFE) in accordance with ASTM D4894; or

e. Other materials as specified (see 6.2) appropriate to the temperature, pressure, and fluid compatibility of the valve. Materials such as glass or graphite filler may be added to material to improve wear resistance.

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3.2.4 Prohibited materials. Valves shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.2.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction. Dilating flow control valves shall consist of three or more rotationally closing plate segments, positioned by gearing. Gearing may be ring, or rack, and pinion. The operating stem shall not enter the flow passage. The valves shall be tight seating at all ports. Valve bodies shall withstand the applied stresses developed during operation of the valve. Valves shall be operable by one person in accordance with MIL-STD-1472. Figure 1 illustrates a typical valve of this construction.

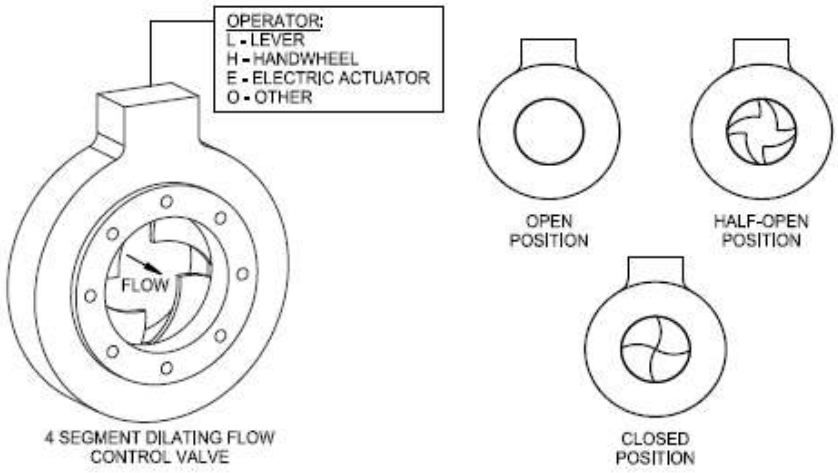


FIGURE 1. Typical dilating flow control valve.

3.3.1 Dimensions. Unless otherwise specified (see 6.2) the valve face-to-face, center-to-face, and end dimensions shall be in accordance with ASME B16.10 and ASME B16.34 requirements for long pattern, steel globe valves, for the valve size specified. Flow passage shall be round bore and shall be sized to provide the minimum Flow factor (Kv) or flow coefficient (Cv) rating as required (see 3.4.8and 6.2). Dimensions and machining of end flanges shall be in accordance with ASME B16.5 of the valve class or class 150 if unspecified. When specified (see 6.2), Navy standard flanges in accordance with Navy Drawing 803-1385892 shall be used. Valve flange bolt holes may be tapped to suit bolting in areas where valve body interferes with through bolting.

3.3.2 Segment motion. Dilating flow control segment motion shall be limited by positive stops at the full-open and full-closed positions. The position shall be shown by permanent and distinct external marks on top of the operating stem. The valve shall incorporate provisions for positively retaining the dilating disk assembly in all positions.

3.3.3 Low friction surfaces. Valve components that rotate shall operate against a low friction seat or seal face material. Low friction treated surfaces shall be either coated or sleeved in accordance with ASTM D4894. Surfaces that rub on one another shall be designed to reduce galling.

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3.3.4 Handwheels and lever handles. Handwheels smaller than 279 millimeters (mm) (11 inches) in diameter and all levers shall be of commercial design and manufactured from steel, ductile iron, or malleable iron or cast manganese bronzes. Handwheels 279 mm in diameter and larger shall be cast steel in accordance with the general dimensions of Navy Drawing 803-1385620. Aluminum or aluminum alloys shall not be used for handwheels or levers. The stem handwheel connection shall be in accordance with Navy Drawing 803-1385620 for all sizes.

3.3.5 Handwheel and lever handle size. Unless otherwise specified (see 6.2), valves under 152 mm (6 inches) shall be furnished either with lever handles or handwheels. Valves 152 mm and larger shall be provided with handwheels operated through enclosed worm and gear mechanisms. Handwheels shall be not less than 152 mm in diameter. Operating forces, nominal lever lengths, and nominal handwheel diameters shall be as specified in table I.

TABLE I. Valve operating torque.

Valve Nominal Size DN [1/] (NPS)	Maximum Allowable Tangential Force [2/] Newtons (pounds-force)	Maximum Operating Torque [3/] Newton-Meters (foot-pounds)	Nominal Lever Length mm (inches)	Nominal Handwheel Diameter mm (inches)
8 (¼)	54.2 (40)	16.3 (12)	127 (5)	178 (7)
25 (1)	54.2 (40)	17.6 (13)	152 (6)	203 (8)
40 (1½)	54.2 (40)	20.3 (15)	178 (7)	230 (9)
50 (2)	54.2 (40)	24.4 (18)	178 (7)	279 (11)
80 (3)	54.2 (40)	27.1 (20)	229 (9)	305 (12)
100 (4)	54.2 (40)	36.6 (27)	279 (11)	406 (16)
150 (6)	54.2 (40)	47.5 (35)	--	533 (21)
200 (8)	54.2 (40)	54.2 (40)	--	610 (24)
250 through 400 (10 through 16)	54.2 (40)	54.2 (40)	--	Up to 914 (36)

NOTES:
1/ Sizes in accordance with ISO 6708.
2/ Measured at the rim of the handwheel.
3/ Measured at valve stem for hand-operated valves and measured at manual-actuator stem for valves with gear boxes.

3.3.6 Manual valve actuators. Worm and gear mechanisms shall be enclosed and shall be commercial items except that they shall be shock and vibration tested while mounted on the valve in an operating condition (see 4.6.1 and 4.6.2).

3.3.7 Remote actuating gear. Valves shall be capable of use with mechanical remote actuating gear as described in S9086-RK-STM-010/505. The adapter for remote actuating gear shall be as specified (see 6.2). Valves shall accept remote actuating gear sized to deliver between 20 and 50 percent more torque than the maximum seat torque listed in table I.

3.3.8 Electric valve actuators. When specified (see 6.2), valves shall be capable of use with electrically-powered valve actuators in accordance with DOD-V-24657 or commercial equivalents. The actuators shall be shock and vibration tested while mounted on the valve in an operating condition (see 4.6.1and 4.6.2).

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3.4 Performance.

3.4.1 Pressure and temperature rating. The valve shall follow standard class rules for pressure-temperature rating in accordance with ASME B16.34. Unless otherwise specified (see 6.2), dilating flow control valves shall be rated to a maximum operating and design of 220 °C (428 °F) and the respective pressure rating of the ASME class.

3.4.2 Working fluid. Dilating flow control valves shall normally serve in water, oil, or gas services. When specified (see 6.2), valves shall serve seawater, sewage, or other working fluids containing high contaminant or solids loads and meet requirements herein.

3.4.3 Shock. Unless otherwise specified (see 6.2), valves shall withstand the grade A, hull mounted, class I, type C shock requirements in accordance with MIL-DTL-901 and MIL-STD-798 (see 4.6.1).

3.4.4 Vibration. Unless otherwise specified (see 6.2), valves shall withstand the type I vibration requirements in accordance with MIL-STD-167-1 (see 4.6.2).

3.4.5 Leak through. Unless otherwise specified (see 6.2), when closed, valves through leakage shall meet class V criteria of ANSI/FCI 70-2.

3.4.6 Hydrostatic withstand. Valves shall withstand an internal hydrostatic test at the pressure specified by ASME B16.34 for its rating, in the open position, for 5 minutes with no external leakage, weeping, or structural distortion (see 4.6.5).

3.4.7 Maintainability. Valve maintenance shall be accomplishable without the need for special training or special tools. Special tools are defined as those tools not listed in the GSA Global Supply Catalog. Valves shall operate without maintenance requirements, except operating mechanism lubrication, for 20,000 open-close operating cycles.

3.4.8 Flow factor. Kv or Cv for valves shall be as specified (see 6.2). Typical flow factors for standard manufacturers trim are noted in 6.5.

3.5 Marking.

3.5.1 Body marking. Valve bodies shall have the pressure rating and manufacturer's name or trademark etched, stamped, or cast integral with the body.

3.5.2 Identification plates. Valves shall bear an identification plate made of corrosion-resisting steel or brass. Identification plates shall be permanently fastened to a part of the valve not subjected to the working pressure. The identification plate shall include the following information:

a. Manufacturer's name or trademark.
b. Size of valve and pressure rating.
c. Body and segments material composition.
d. Manual valve actuator mechanical advantage (if applicable).
e. Manufacturer's identification number or drawing number.
f. Specification number "MIL-DTL-32632".
g. PIN (see 1.3).

3.6 Workmanship. Valves shall be thoroughly cleaned both inside and out. Valves shall be free from burrs, loose sand, grit, scale, metal shavings, and all other contaminants that can affect performance and damage sealing surfaces. There shall be no sharp edges. All sealing, rubbing, and coated surfaces shall be smooth with no blemishes, scratches, scrapes, or gouges. Machined surfaces shall not exceed 125 roughness height rating (RHR) as defined in ASME B46.1, unless a rougher finish is allowed by specifications. Valve bodies shall meet the visual acceptance criteria for castings in accordance with S9074-AR-GIB-010/278.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one valve of each class when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.5 and the tests specified in table III Acceptance of the inspected sizes shall constitute acceptance of the sizes not inspected as specified in table III

TABLE II. First article approval sizes.

Size tested DN (NPS)	**Approves these sizes DN (NPS)**
8 (¼)	8 (¼)
50 (2)	25 (1), 40 (1½), 50 (2)
150 (6)	80 (3), 100 (4), 150 (6)
250 (10)	200 (8), 250 (10)
300 (12)	300 (12)
350 (14)	350 (14)
400 (16)	400 (16)

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests specified in table III. Every valve offered to the Government shall be tested in accordance with 4.6.4. Conformance inspection shall be performed on samples selected in accordance with 4.4 or as specified (see 6.2).

4.4 Sampling.

4.4.1 Sampling for examinations and tests. A random sample of valves shall be selected from each lot offered to the Government in accordance with ANSI/ASQ Z1.4 at inspection level II and the Acceptance Quality Limit (AQL) specified (see 6.2).

4.4.2 Lot. A lot shall consist of all valves of the same class and size produced under the same conditions and offered for delivery at one time, as specified. A sample valve shall consist of one completely assembled dilating flow control valve.

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TABLE III. Examination and tests.

Inspection	Requirement	First Article Inspection	Conformance Inspection
Materials	3.2	4.5.1	--
Visual and dimensional examination	3.3, 3.4.1, 3.4.2	4.5.2	4.5.2
Examination after first article tests	3.3	4.5.3	--
Shock	3.4.3	4.6.1	--
Vibration	3.4.4	4.6.2	--
Leak through	3.4.5	4.6.3	--
Operational	3.4.6	4.6.4, 4.6.5	--
Hydrostatic	3.4.6	4.6.5	4.6.5
Maintainability	3.4.7	4.5.3	--
Flow factor	3.4.8	4.6.6	--
Markings	3.5	4.5.2	4.5.2
Workmanship	3.6	4.5.2	4.5.2

4.5 Examinations.

4.5.1 Examination of materials. Materials used in the manufacture of valves shall be examined to determine conformance to 3.2 as applicable (see 6.2).

4.5.2 Visual and dimensional examination. Sample valves selected shall be visually and dimensionally examined to verify compliance with the nonoperational requirements of this specification and the manufacturer's drawings. Each of the sample valves selected shall be examined in accordance with table III and table IV. Any valve in the sample containing one or more defects shall be rejected, and if the number of defective valves in any sample exceeds the acceptable number for that sample, the lot represented by the sample shall be rejected.

4.5.3 Examination after first article tests. Upon conclusion of first article tests, the valve shall be disassembled and visually and dimensionally examined for damage and wear. Any valve or sample lot not meeting defect criteria of table IV shall be rejected. Disassembly and reassembly shall be performed to determine the practicability of maintaining a valve of this design for shipboard use. A maintainability demonstration shall be conducted by removing and replacing elastomers, segments, springs and operating gear. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used. Particular attention shall be given to avoid the possible loss of small parts by the maintenance crew.

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TABLE IV. Classification of defects.

Categories	Major Defects
101	Class not as specified (see 1.2)
102	Material not as specified (see 3.2)
103	Construction not as specified (see 3.3.1)
104	Shock resistance not as specified (see 3.4.3)
105	Vibration resistance not as specified (see 3.4.4)
106	Leak through not as specified (see 3.4.53.6)
107	External leakage not as specified (see 3.4.6)
108	Maintainability not as specified (see 3.4.7)
109	Flow factor not as specified (see 3.4.8)
110	Body marking and identification plates not as specified (see 3.5)
111	Workmanship not as specified (see 3.6)

4.6 Tests.

4.6.1 Shock. The valve shall be subjected to the grade A shock test in accordance with MIL-DTL-901 to determine conformance to 3.4.3. Valves shall be hydrostatically pressurized internally to the maximum gauge pressure as specified in 3.4.1. Testing medium shall be clean tap water at temperatures not less than 4 °C (40 °F) or exceeding 52 °C (125 °F). Medium may not contain additives except corrosion inhibitors that do not increase the viscosity or surface tension of the water, and residual cutting oil from machining. Momentary malfunction at the time of impact blow is permitted and acceptable. Permanent deformation, misalignment, or functional impairments shall be cause for rejection. Permissible seat leakage shall be as specified in 3.4.5.

4.6.2 Vibration. The valve shall be subjected to the type I vibration test in accordance with MIL-STD-167-1 to determine conformance to 3.4.4.

4.6.3 Leak through and seat leakage. The valve shall be tested in accordance with ANSI/FCI 70-2 and shall meet class V criteria or better (see 3.4.5).

4.6.4 Operational tests. Operational tests shall be performed on each valve size and class. Operational tests shall be conducted in accordance with the test conditions of 4.6.5.1, except if otherwise specified (see 6.2), for seawater, sewage, or other working fluid which will contain high contaminant or solids loads. In these cases, the test fluid shall be the expected working fluid. Tests shall alternately apply the pressure specified in 3.4.1 with the pressure on the seat with the valve in the shut position for 20,000 cycles. One cycle shall consist of the above specified pressure applied in the shut position and in the open position. At the completion of the operational test, the external leakage test of 4.6.5.2 shall be performed.

4.6.4.1 Operating torque test following a period of non-operation. A differential pressure equal to the rated pressure shall be maintained across the valve for 40 hours. After this period of time has elapsed, the valve shall be fully cycled (closed to open to closed) every 2 hours and the breakaway torque recorded. This shall be continued until such time as three consecutive readings that are both within ±5 percent of each other are obtained. Any torque measurement greater than the maximum operating torque (see table I) shall be cause for rejection.

4.6.5 Hydrostatic withstand pressure and external leakage. Every valve offered to the Government shall be tested in accordance with 4.6.5.1 and 4.6.5.2 after being fully cycled (opened and closed) to demonstrate acceptable assembly and operation.

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4.6.5.1 Test conditions. Each valve shall be hydrostatically tested for structural integrity and seat leakage using clean tap water at temperatures not less than 4 °C (40 °F) and not more than 66 °C (150 °F) and shall contain no additives other than cutting oil. Test fixtures shall be adjusted to permit valve body expansion and elongation due to internal pressure to the fullest extent possible. The thrust on the ends shall be the minimum required to facilitate testing and prevent leakage to the atmosphere.

4.6.5.2 External leakage hydrostatic test. Each valve shall be subjected to the test pressure, as specified by ASME B16.34, in the open position for 5 minutes. Any external weeping or leakage during or after pressurizing or permanent deformation after pressurizing shall be cause for rejection.

4.6.6 Flow factor. Flow factor (coefficient) shall be verified using the methods and criteria of ANSI/ISA 75.02.

4.7 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.4 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Valves covered by this specification are high-reliability, are intended for service aboard Navy ships with a 30-year service life, and meet Navy shock and other unique requirements.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Valve classification, end type, size, operator to be used, and potable water suitability (see 1.2 and 4.4.2).

c. When first article inspection is required (see 3.1).

d. Materials and fasteners requirements if other than specified (see 3.2, 3.2.1, 3.2.1.f, 3.2.2, 3.2.3, and 3.2.3.e).

e. Requirement for prohibited materials verification (see 3.2.4 and 4.7).

f. When valve face-to-face, center-to-face, and end-to-end dimensions are to be in accordance with ASME B16.10 and ASME B16.34 long pattern, steel globe valve long pattern (see 3.3.1).

g. If Navy standard flange is required (see 3.3.1).

h. Lever handle or handwheel requirements if other than specified (see 3.3.5).

i. Remote actuating gear adapter requirements (see 3.3.7).

j. If valves are required to be capable of use with electric valve actuators (see 3.3.8).

k. Valve rating if other than specified (see 3.4.1).

l. Expected working fluid, if containing high contaminant or solids loads, such as seawater or sewage (see 3.4.2 and 4.6.4).

m. Shock requirements if other than specified (see 3.4.3).

n. Vibration requirements if other than specified (see 3.4.4).

o. Leak through requirement if other than specified (see 3.4.5).

p. Flow factor requirement if other than specified (see 3.4.8).

q. Additional conformance inspections if required (see 4.3).

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r. AQL of 1.0 (see 4.4.1).

s. Specific materials testing requirements if required (see 4.5.1).

t. Operational test fluid if not clean tap water (see 4.6.4).

u. Packaging requirements (see 5.1).

v. Documentation if required (see 6.4).

6.3 First article. When first article inspection is required (see 6.2), the contracting officer should provide specific guidance to offerors that the item is a first article sample (see 3.1) and the number of items to be tested as specified in 4.4. Acquisition documents should include specific instruction regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternative bids unless specifically requested to do so in the solicitation.

6.4 Recommended documentation. The following list of documentation is recommended to be requested as part of the acquisition of any valve under this specification:

a. Engineering drawings.

b. Materials listing.

c. Installation drawings.

d. Technical manual.

e. Test reports.

f. Certifications.

g. Onboard repair parts list.

6.5 Typical flow factors. Typical flow factors are as specified in table V.

TABLE V. Typical flow factors.

DN Size	NPS Size	Minimum Flow Factor	
		Kv	Cv
8	¼	4.3	5
25	1	17	20
40	1½	46	55
50	2	91	105
80	3	160	185
100	4	323	375
150	6	1164	1350
200	8	2414	2800
250	10	3707	4300
300	12	5732	6650
350	14	6594	7650
400	16	8448	9800

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6.6 Subject term (key word) listing.

Flow Coefficient

Flow Factor

Iris

Petal

Segment

Shutter

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 69

Review activities:
Army – MI
Navy – AS
Air Force – 03, 71
DLA – CC

Civil agency:
GSA – FAS

Preparing activity:
Navy – SH
(Project 4820-2019-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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